

CSE 214

Online on Creational Pattern

Time: 30 minutes

You are designing software for a customized bicycle manufacturer. A bicycle has three major components: a **Frame**, a **Gear System**, and a **Tire Type**.

The factory currently produces two standard models:

1. **The Commuter:** Features "Aluminum Frame", "Single Speed Gear", and "Road Tires".
2. **The Mountain Beast:** Features "Carbon Fiber Frame", "12-Speed Gear", and "Off-road Grip Tires".

Task: Implement a solution to construct the **Bicycle** object. The creation process should be step-by-step (e.g., build frame, then add gears, then add tires). You must create a class that directs the construction process so that the client only needs to ask for a specific model to get the fully assembled product.

- Use Strings to represent the components (no need for complex component classes).